

FACULTY OF ENGINEERING

EXAM ASSIGNMENT

INGB413

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Table of Contents

1. Introduction & Background	3
2. Literature Review	4
3. Data Collection	6
4. Data Analysis	9
4.1. Data Distribution	9
4.2. Statistical Process Control (SPC)	10
4.3. Hypothesis Testing	12
4.4. Capability	14
4.5. Profitability	14
5. Conclusion	15
References	16
Appendix A (Process Mapping)	17
Process 1 Mapping	17
Process 2 Mapping	18
Appendix B (Data)	19

List of Figures

Figure 1.1: Tangram Puzzle in Wooden Board.....	3
Figure 3.1: Process 1 mapping	6
Figure 3.2: Process 2 mapping	6
Figure 3.3: Boxplot of Process 1	7
Figure 3.4: Boxplot of Process 2	7
Figure 4.1: Process 1 Observed vs Binomial Distribution	9
Figure 4.2: Process 2 Observed vs Geometric Distribution	10
Figure 4.3: P-chart for Process 1	11
Figure 4.4: G-chart for Process 2	12

List of Tables

Table 3.1: Collected Data Summary.....	7
Table 4.1: Chi-Square Test Summary	9
Table 4.2: Summary of P Control Chart	10
Table 4.3: Summary of G Control Chart	11
Table 4.4: DPMO Calculations	14
Table 4.5: Cost Summary	14

1. Introduction & Background

Tangram puzzles are traditional Chinese puzzles comprising of seven geometric pieces that can be rearranged to form a wide range of distinct shapes. *Tangra Inc.* is a company specialising in the production of these puzzles. The company sources individual puzzle pieces from various suppliers, assembles them into a complete set within a wooden board and then ships the final product to customers. All pieces are distinct from one another by ways of their colour and/or shape and fit perfectly into the wooden board. The fully assembled tangram, enclosed in the wooden board, is depicted in figure 1.1 below.



Figure 1.1: Tangram Puzzle in Wooden Board

If even just one piece of the tangram puzzle above deviates a bit from the desired shape, the puzzle will not fit in the wooden board and cannot be shipped to customers. This piece is then considered as defective. The production process is required to run at a 2-sigma level as specified by *Tangra Inc.*

A recent increase in defective puzzle pieces on the assembly lines has been observed and current demand cannot be met. It is decided to conduct two experiments (Process 1 & Process 2) and to use the data gathered from these experiments to investigate and highlight what the findings of the two processes in terms of quality parameters are. The effect of these quality defects on the profitability of the business, will also be examined.

2. Literature Review

Effective quality management ensures that processes consistently produce defect-free outputs, reduces variation and supports business goals such as customer satisfaction and profitability (ISO, 2015). The application of quality management principles in production environments enables organisations to identify root-causes of variation and implement process controls to maintain stable and predictable outputs.

Prevention costs, appraisal costs, internal failure costs and external failure costs are the four categories into which the Cost of Quality (CoQ) framework divides quality-related costs (Kau & Nel, 2019). Process 1, sampling and defect removal prior to assembly, is more in line with control costs (prevention and appraisal costs). As assemblies with flaws are discarded, Process 2 results in higher internal failure costs. Determining each process's position within the CoQ model offers information about the financial effects of decisions pertaining to quality.

A trade-off between sampling inspection and full inspection can be identified. Sampling is less costly and often sufficient when incoming quality levels are predictable, whereas full inspection or detection during operation may be necessary in high-risk environments (Evans & Lindsay, 2017). Process 1 utilises a small and fixed sample ($n=5$) for inspection, aiming to detect defects early. In contrast, Process 2 reacts to defects during the assembly process, often resulting in wasted material and time.

The application of Statistical Process Control (SPC) techniques is fundamental in monitoring process behaviour and identifying the presence of assignable variation in manufacturing or assembly processes (Montgomery, 2013). One of the primary tools in SPC is the control chart, which allows for the visualisation of process performance over time. Control charts are especially useful for monitoring process stability and guiding data-driven decisions. This is in alignment with the ISO 9001:2015 emphasis on evidence-based quality management (ISO, 2015).

In this report, two different control charts are employed based on the nature of the data generated by each tangram assembly process. For Process 1, the p-chart is used as it effectively tracks the proportion of defective units in a sample. This process involves taking a fixed sample size of five items before assembly and identifying how many are defective. Since each part is defective or not, this results in binomial (attribute) data (NIST, 2010). The p-chart is thus most suitable for monitoring the proportion of nonconforming parts across repeated samples. Charts like the np-chart, which plots the number of defectives instead of the proportion, are less appropriate for small or variable sample sizes. Similarly, c- and u-charts are not used because they assume a single item can contain multiple defects, which is not the case in this scenario (Montgomery, 2013).

For Process 2, the g-chart is selected due to the data collection method that records the number of non-defective items placed before encountering a defective one. This aligns with the geometric distribution, where the probability of success continues until the first failure occurs. In this case, when a defective part is identified, the entire tangram assembly is discarded, and the number of items placed up until that point is counted. The g-chart is particularly effective when sample sizes are not fixed, as it does not require a predefined number of items in each observation. This separates it from p-charts, which rely on a consistent or known sample size to calculate proportions. Other control charts, such as c- or u-charts are also not appropriate for use in this case, because they assume multiple defects per unit or per sample, which does not match the data collection method used in Process 2 (Evans & Lindsay, 2017).

Alongside SPC, this report utilises hypothesis testing to perform a structured comparison between the two processes. Hypothesis testing is a statistical method used to draw inferences about population parameters based on sample data (Montgomery, 2013). For this report a two-sided z-test is the most appropriate as the population standard deviations are unknown, and sample means are used to compare process quality.

Process capability refers to the ability of a process to produce outputs that meet specified quality standards or customer requirements. It is commonly evaluated using statistical measures such as the process capability index (C_p , C_{pk}) or sigma levels, which indicate how well the process performs relative to its specification limits. In the context of attribute data, capability can also be assessed using defect rates, where emphasis is put on sigma levels that are translated into defects per million opportunities (DPMO). A 2-sigma process yields approximately 40 400 defects per million units, indicating a relatively low capability compared to higher sigma levels (Kumar *et al.*, 2018). This form of analysis is important for identifying whether a process is capable of consistently delivering quality outputs.

The literature highlights that quality management relies on a combination of preventive strategies, statistical control, capability analysis and cost-conscious decision-making. The use of SPC tools like control charts, combined with hypothesis testing, capability analysis and cost of quality analysis provides a solid framework to evaluate and improve process performance.

3. Data Collection

In order to investigate the problem, two processes are tested. These processes work as follow:

Process 1

- Receive raw material in bulk
- Take samples of 5 and test for defects
- Remove all defective parts
- Assemble tangrams

Process 2

- Receive raw material already sorted
- Pick parts from the sorted bins
- Assemble tangrams
- If any defects are found, the entire assembly is discarded (wooden board & puzzle pieces)
- Number of puzzle pieces already placed and the number of defective pieces in each discarded assembly is counted

Both tests are conducted for 15 minutes each. Within those 15 minutes, all of the puzzle pieces of process 1 was checked for defects and sorted. For process 1, the sample size for each sample is 5, but the sample size of each sample in process 2 differs and is dependent on the amount of puzzle pieces placed on the board up until the point at which the first defect is found. Process 2 can however take less than 15 minutes as it will be stopped once one of the type of puzzle pieces are finished. A defective piece can be described as a piece that does not fit the space allocated to it or hinders other pieces to fit in their allocated spaces. The process maps of the two processes can be seen in figures 3.1 and 3.2 below. A larger scale depiction of the process mappings can be seen in Appendix A.

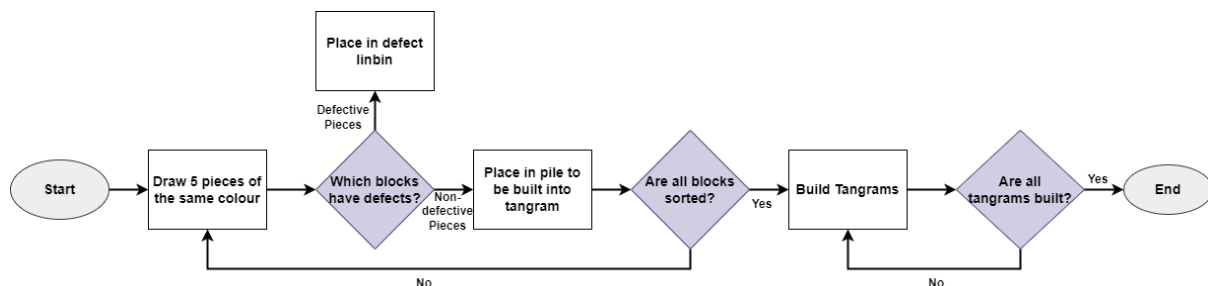


Figure 3.1: Process 1 mapping

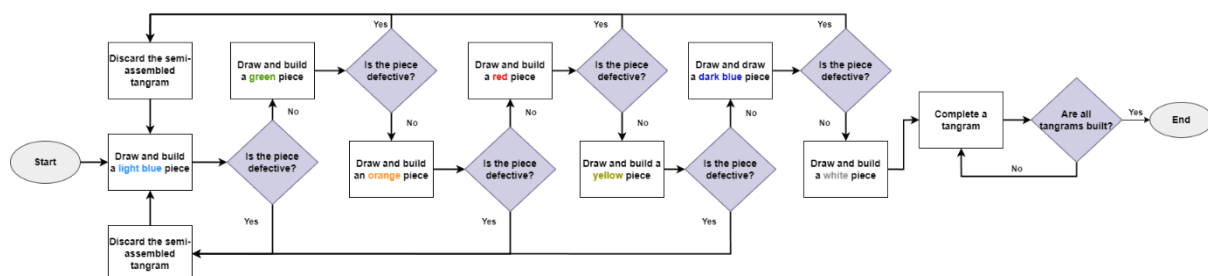


Figure 3.2: Process 2 mapping

For process 2, the order in which the puzzle pieces were selected each time was determined by the proportion defects of each piece from process 1. The pieces with the lowest proportion

defects (light blue & green) were selected first, continuing until the pieces with the highest proportion defects are selected (white) or if a defect is found, discarding the semi-assembled tangram and starting at the light blue pieces again.

Below, in table 3.1, is a summary of the data collected in processes 1 and 2. The full sheets of data can be seen in Appendix B.

Table 3.1: Collected Data Summary

Item	Process 1	Process 2
Samples (m)	42	30
Sample Size (n)	5	3-6
Tangrams Completed	0	0
Time Taken	15min	7.5min

Boxplots are widely recognised as effective tools for summarising and visualising data distributions. They concisely represent key statistical measures and provide a clear depiction of a dataset's tendencies, variability and potential outliers. This makes boxplots particularly useful for comparing distributions across different groups or categories (Iglewicz, 2011). The boxplots for the two processes are plotted using R and are depicted in figures 3.3 and 3.4 below.

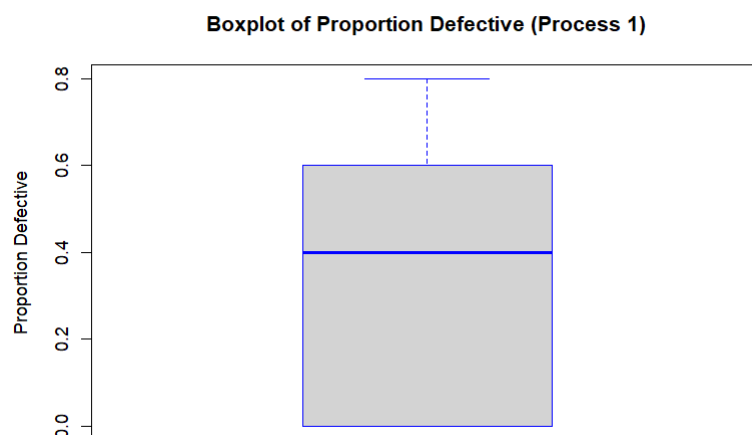


Figure 3.3: Boxplot of Process 1

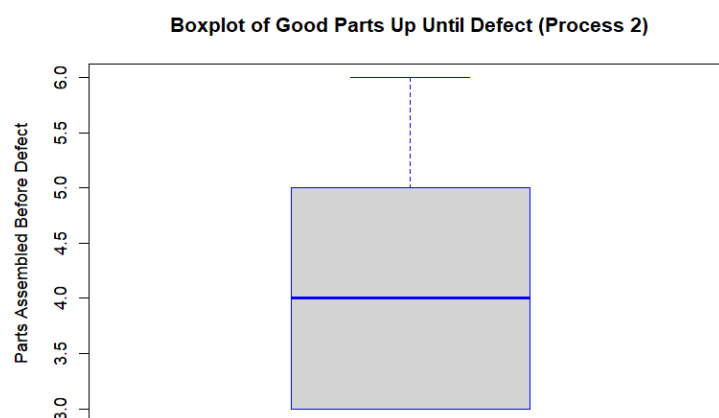


Figure 3.4: Boxplot of Process 2

The first boxplot (figure 3.3) represents the proportion of defective items in Process 1. The median proportion defective is 0.4, indicating that nearly half of the items in a typical sample are defective. The interquartile range (IQR) spans from 0 to 0.6, suggesting a considerable spread in the data. The lower whisker at zero implies that some samples had no defects at all, while the upper whisker extends to 0.8, showing that defect rates can be quite high in some cases. Overall, the distribution is relatively symmetric but slightly skewed toward higher defect rates. This points to a process that is highly variable and frequently produces a significant number of defects, indicating poor quality control.

The second boxplot (figure 3.4) shows the number of good parts assembled up until a defect occurs in Process 2. The median is 4, meaning that on average, four good parts are produced before a defect is encountered. The IQR ranges from 3 to 5, reflecting a narrower spread and, therefore, more consistent process performance. The whiskers extend from 3 to 6, emphasising the idea that most of the data falls within a tight range. This boxplot suggests a process that is relatively stable and predictable.

4. Data Analysis

4.1. Data Distribution

The Chi-square goodness-of-fit test is a statistical method used to evaluate whether observed categorical data conform to an expected theoretical distribution. The test is particularly useful when analysing count data across discrete categories, such as the number of defects or occurrences within fixed intervals. It works by comparing the observed frequencies with the expected frequencies derived under a specified probability model, such as the binomial or Poisson distribution. A large difference between the observed and expected values leads to a high chi-square statistic and a low p-value, indicating poor fit. Conversely, a small chi-square statistic and high p-value suggest that the data do not significantly differ from the expected distribution (Agresti, 2013). The proposed data distributions, Chi-squared test statistic, Chi-square values, Degrees of Freedom and p-values of the two processes are summarised in table 4.1 below.

Table 4.1: Chi-Square Test Summary

Item	Process 1	Process 2
Data Distribution	Binomial	Geometric
Chi-square Test Statistic	$\chi^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}$	
Chi-square Value	2.584	1.8003
Degrees of Freedom	2	3
p-value	0.2747	0.6149

Several distributions were evaluated for process 1 and none provided a satisfactory fit based on visual inspection and goodness-of-fit tests (p-values < 0.05). This suggests that the process variability does not follow a standard distribution and will benefit from grouping defect levels into bins. To further validate the above data distributions, the observed data (blue) is plotted against the expected distribution (red) for the two processes in figures 4.1 and 4.2 below, respectively.

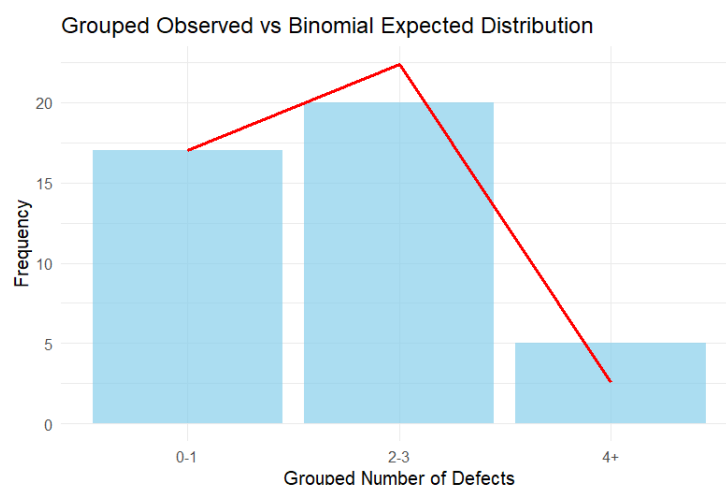


Figure 4.1: Process 1 Observed vs Binomial Distribution

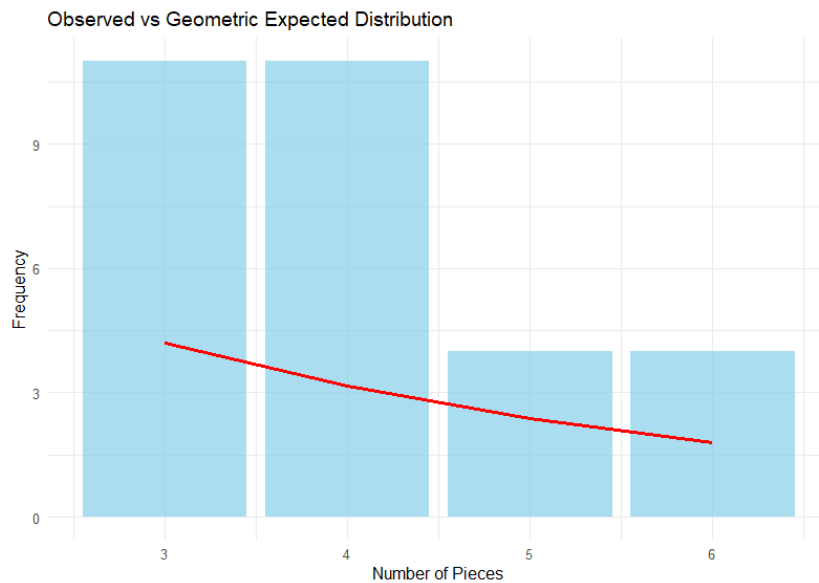


Figure 4.2: Process 2 Observed vs Geometric Distribution

4.2. Statistical Process Control (SPC)

From the above literature study, it is concluded that a p-chart and g-chart will be constructed for process 1 and process 2 respectively. The charts will be utilised to determine whether the process is in statistical control based. A table with the formulas and values (rounded to five decimals) used to construct the control charts can be seen below in table 4.2. The Upper and Lower Control Limits (UCL & LCL) are three-sigma levels away from the centre line (mean), while the Upper and Lower Warning Limits (UWL & LWL) are two sigma levels away from the centre line. In the table below, m refers to the amount of samples drawn and n to the sample size.

Table 4.2: Summary of P Control Chart

Item	Formula	Calculated Value
Centre Line	$\bar{p} = \frac{\text{proportion of defects}}{m}$	0.36190
UCL	$\bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$	1.00663
LCL	$\bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$	-0.28282
UWL	$\bar{p} + 2 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$	0.79172
LWL	$\bar{p} - 2 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$	-0.06791

Since control charts for attribute data (like p-charts and np-charts) monitor proportions or counts of defective items, which cannot be negative by definition, the LCL and LWL are truncated at zero (Montgomery, 2013). The p-chart for process 1 is depicted in figures 4.3 below.

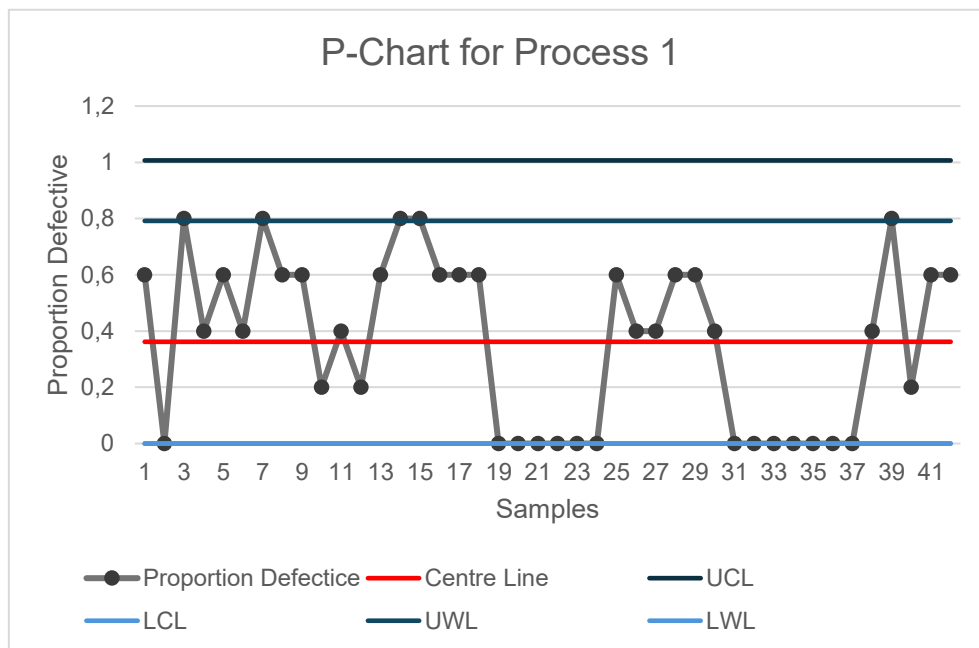


Figure 4.3: P-chart for Process 1

When taking the Western Electric Handbook's decision rules for detecting non-random patterns on control chart, we can see that the process depicted in the chart above is out of control. This is evident where between samples 13-15, both samples 14 and 15 plot beyond two-sigma warning limits.

A g-control chart is a type of control chart used in SPC to monitor the number of opportunities between rare events or nonconforming items. It is especially useful when defects or nonconformities are infrequent, such as monitoring failures. The "g" stands for the geometric distribution, which coincides with the data distribution identified for process 2 earlier in this report. A summary of the formulas and calculated values (rounded to 4 decimals) use to plot the g-chart can be seen in table 4.3, below. Since process 2 has a geometric distribution, the centre line is taken to be 0.693 times the average. The subsequent control chart is also depicted below in figure 4.4.

Table 4.3: Summary of G Control Chart

Item	Formula	Calculated Value
Average pieces placed	$\bar{g} = \frac{\text{total no. of pieces}}{\text{samples}}$	4.0333
Centre Line	$0.693\bar{g}$	2.7951
UCL	$\bar{g} + 3\sqrt{\bar{g}(\bar{g} + 1)}$	17.5504
LCL	$\bar{g} - 3\sqrt{\bar{g}(\bar{g} + 1)}$	-9.4837

UWL	$\bar{g} + 2\sqrt{\bar{g}(\bar{g} + 1)}$	13.0447
LWL	$\bar{g} - 2\sqrt{\bar{g}(\bar{g} + 1)}$	-4.9780
U1S	$\bar{g} + \sqrt{\bar{g}(\bar{g} + 1)}$	8.5390
L1S	$\bar{g} - \sqrt{\bar{g}(\bar{g} + 1)}$	-0.4723

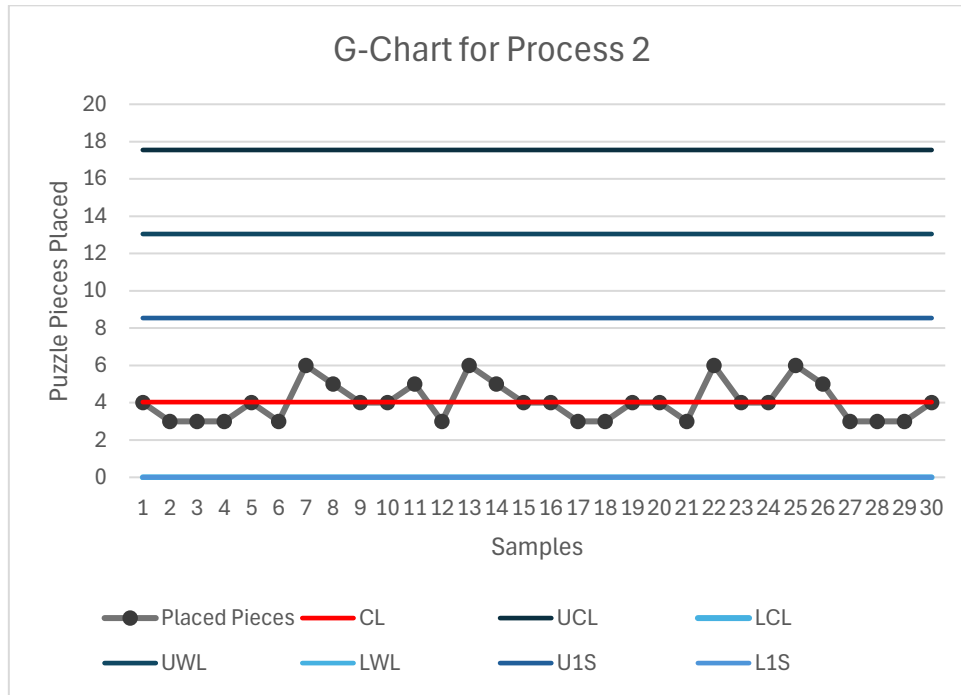


Figure 4.4: G-chart for Process 2

The LWL and LCL are also truncated to zero in the g-chart below. Also taking into account the Western Electric Decision rules, it is evident that the process 2 is deemed to be in statistical control and no outliers are present.

Even though process 2 is in control, customer specifications are not met since no tangram could be completed before the test batch of puzzle pieces were finished. Process 1 being out of control indicates that the puzzle pieces received from the suppliers are unpredictable.

4.3. Hypothesis Testing

In order to validate whether the number of defects found during the processes falls within 2-sigma control limits, a statistical process control (SPC) check will be done for process 1 by means of hypothesis testing. The 2-sigma value (0.79172) in table 4.2 above is used. The test is formulated as follows,

$$H_0: p_1 \in [0, 0.79172]$$

$$H_1: p_1 > 0.79172$$

The test statistic is calculated by taking the sample with the most defects.

$$Z_0 = \frac{p - \bar{p}}{\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}}$$

$$Z_0 = \frac{0.8 - 0.3619}{\sqrt{\frac{0.3619(1 - 0.3619)}{5}}}$$

$$|Z_0| = 2.0385$$

As this is a 2-sigma test, the null hypothesis is rejected if $|Z_0| > 2$ and can from the above test statistic thus be rejected, thus meaning the process is out of control as suggested earlier in this report.

An SPC check will be done for process 2 by means of a mean z-test hypothesis test. The 2-sigma value (13.0447) in table 4.3 above is used. The test is formulated as follows,

$$H_0: p_1 = 13.0447$$

$$H_1: p_1 > 13.0447$$

The z-test statistic is calculated as follows,

$$Z_0 = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{k}}}$$

$$Z_0 = \frac{4.0333 - 13.0447}{\frac{4.5057}{\sqrt{30}}}$$

$$|Z_0| = 10.9544$$

As $\alpha = 0.05$, it is found that $Z_{0.05} = 1.645$. From this it is concluded that since $|Z_0| > Z_{0.05}$, the null hypothesis is rejected and the process is in control. This coincides with the findings in section 4.2 above.

To further determine which process between the two yields the least defects, a two-proportion z-test is conducted in R to identify if there is any significant difference in the defect proportions between the two processes, regardless of direction.

The hypothesis test is formulated as:

$$H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2$$

The test statistic is calculated as follows,

$$Z_0 = \frac{p_1 - p_2}{\sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$Z_0 = \frac{0.361905 - 0.247934}{\sqrt{0.320242(1 - 0.320242)\left(\frac{1}{210} + \frac{1}{121}\right)}}$$

$$|Z_0| = 2.140258$$

As this is a two-tailed test and $\alpha/2 = 0.025$ is used, it is found that $Z_{0.025} = 1.96$ and the p-value is found to be 0.03233. From this it is concluded that since $|Z_0| > Z_{0.025}$ and $0.03233 \leq 0.05$, the null hypothesis is rejected and process 1 has fewer defects than process 2. The 95% confidence interval calculated in R of [0.01325315, 0.21468860] further suggest that process 1 has between 1.3% and 21.4% less defects than process 2.

4.4. Capability

The process capability analysis with attribute data is done to further confirm if the processes comply with the 2-sigma level that *Tangra Inc.* specifies. If the process operates at a 2-sigma level, it is expected that 95.96% of units fall within 2-sigma of the mean. This translates to 40 400 parts per million parts that are defective. The DPMO calculations is in table 4.4 below.

Table 4.4: DPMO Calculations

Item	Process 1	Process 2
Formula	$\frac{\text{Total no. of defects}}{\text{No. of units} \times \text{No. of opportunities}}$	
DPMO	$\frac{76}{5 \times 42} \times 1\,000\,000 = 361\,905$	$\frac{30}{121} \times 1\,000\,000 = 247\,934$

From the table above, it is evident that not one of the two processes comply with the 2-sigma level specified by *Tangra Inc.*

4.5. Profitability

The costs of raw materials and defects are given and by using these costs a comparison based on the two experiments conducted can be done in table 4.5 below.

Table 4.5: Cost Summary

Item	Value/item	Process 1	Process 2
Puzzle Piece	-R2,00	-R156	-
Wooden Board	-R5,00	-R0,00	-
Defective semi-assembled unit	-R20,00	-	R600,00
Completed product sold	R25,00	R0,00	R0,00
Profit/Loss	-	-R156,00	-R600,00

Both processes experienced a loss as is, but process 1 experienced much less of a loss than process 2. This indicates that in regards to profitability, process 1 is much better to implement than process 2. If more time was allocated to process 1, no loss would have been made since all subsequent tangrams would have been completed and sold for R25,00 each, making up for the loss. Taking the data from process 1, it can be calculated that a total of 10 tangrams (R250,00) could have been completed with all the pieces sorted in the 15 minutes, subsequently resulting in a profit of R94,00.

5. Conclusion

This report provides a comprehensive evaluation of two tangram assembly processes by analysing their quality performance through statistical and process capability techniques. The data distributions, binomial for process 1 and geometric for process 2, were identified in order to select appropriate control chart types. This allows for effective monitoring of process stability. Hypothesis testing was conducted to determine whether the defect rates observed in each process fell within acceptable statistical limits, specifically the 2-sigma threshold as specified by *Tangra Inc.*, as well as to determine which process yields the best results. Process capability analysis, using DPMO and 2-sigma level benchmarks, revealed that both process 1 and process 2 exceeded acceptable defect levels, subsequently indicating a need for quality improvement. Finally, the profitability implications were assessed and showed that process 1 is by far more profitable than process 2.

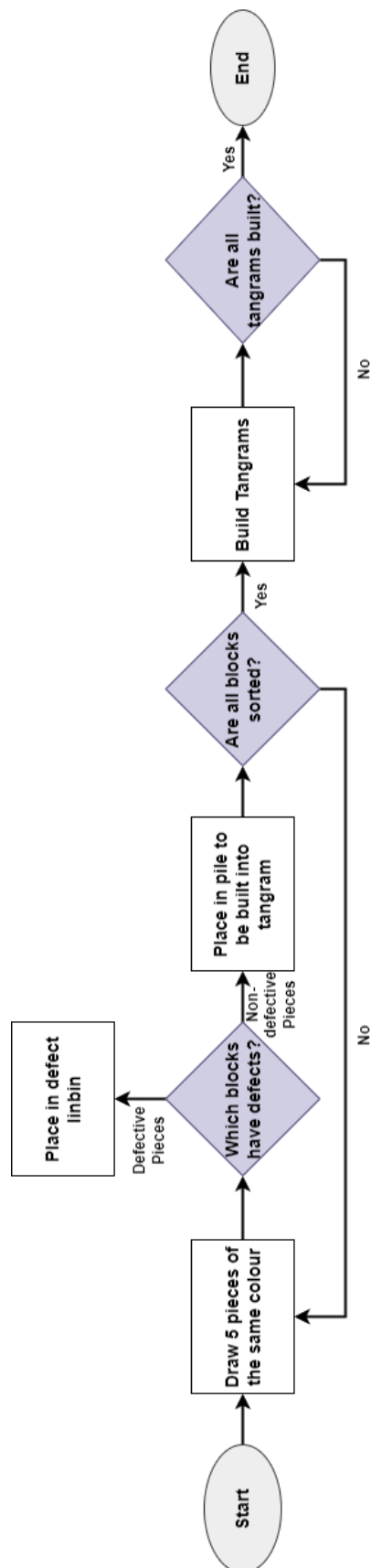
Overall, it is suggested that process 1 should be implemented rather than process 2 as it is more profitable. All puzzle pieces received in bulk must thus first be sorted and defective pieces removed before tangram assembly starts. However, process 1 does not comply with the 2-sigma specification, which suggests that further investigation into the implementation of other processes must be done or the choice of supplier must be reevaluated. This is because almost two-thirds of product has to be scrapped because of defects in the pieces received from the supplier.

References

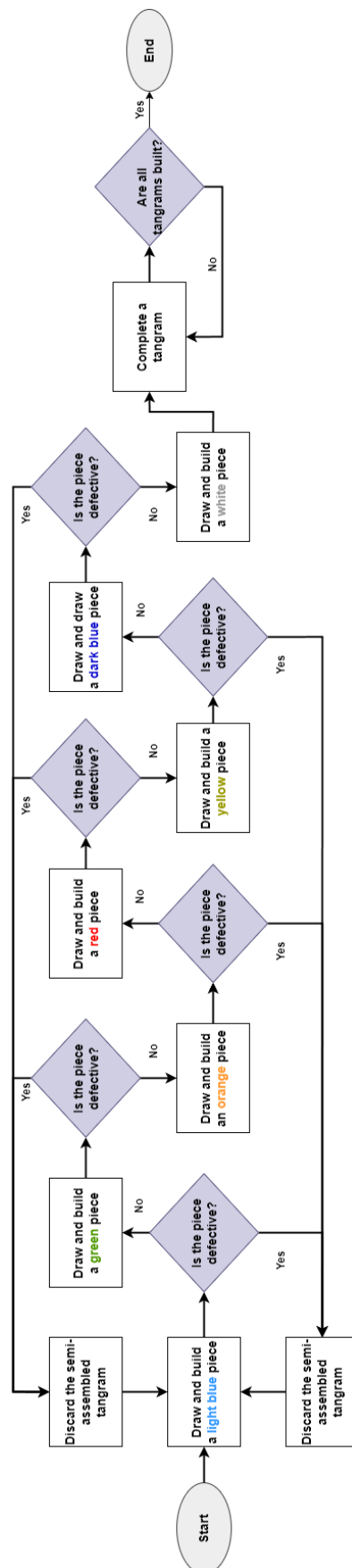
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Appendix A (Process Mapping)

Process 1 Mapping



Process 2 Mapping



Appendix B (Data)

PROCESS 1					
Sample	Colour	Defects	Sample	Colour	Defects
1	Red	3	25	Dark Blue	3
2	Red	0	26	Dark Blue	2
3	Red	4	27	Dark Blue	2
4	Red	2	28	Dark Blue	3
5	Red	3	29	Dark Blue	3
6	Red	2	30	Dark Blue	2
7	Yellow	4	31	Green	0
8	Yellow	3	32	Green	0
9	Yellow	3	33	Green	0
10	Yellow	1	34	Green	0
11	Yellow	2	35	Green	0
12	Yellow	1	36	Green	0
13	White	3	37	Orange	0
14	White	4	38	Orange	2
15	White	4	39	Orange	4
16	White	3	40	Orange	1
17	White	3	41	Orange	3
18	White	3	42	Orange	3
19	Light Blue	0			
20	Light Blue	0			
21	Light Blue	0			
22	Light Blue	0			
23	Light Blue	0			
24	Light Blue	0			

Defect

x

Discarded

0

Defect	x
Discarded	0

PROCESS 2							
Sample	Light Blue	Green	Orange	Red	Yellow	Dark Blue	White
1	0	0	0	X			
2	0	0	X				
3	0	0	X				
4	0	0	X				
5	0	0	0	X			
6	0	0	X				
7	0	0	0	0	0	X	
8	0	0	0	0	X		
9	0	0	0	X			
10	0	0	0	X			
11	0	0	0	0	X		
12	0	0	X				
13	0	0	0	0	0	X	
14	0	0	0	0	X		
15	0	0	0	X			
16	0	0	0	X			
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24	0	0	0	X			
25	0	0	0	0	0	X	
26	0	0	0	0	X		
27	0	0	X				
28	0	0	X				
29	0	0	X				
30	0	0	0	X			